Graphs

Wednesday, October 29, 2025 3:01 PM

$G = (V, E)$, $H = (W, F)$

Subgraph: $W \subseteq V$, $F \subseteq E$

Induced subgraph: $W \subseteq V$, $F = \{xy \in E : x, y \in W\}$ (includes all edges of same vertices)

Spanning subgraph: $W = V$ (includes all vertices)

Theorem: $\sum_{v \in V} \deg_G(v) = 2|E|$ (sum of degree = 2x # of edges)

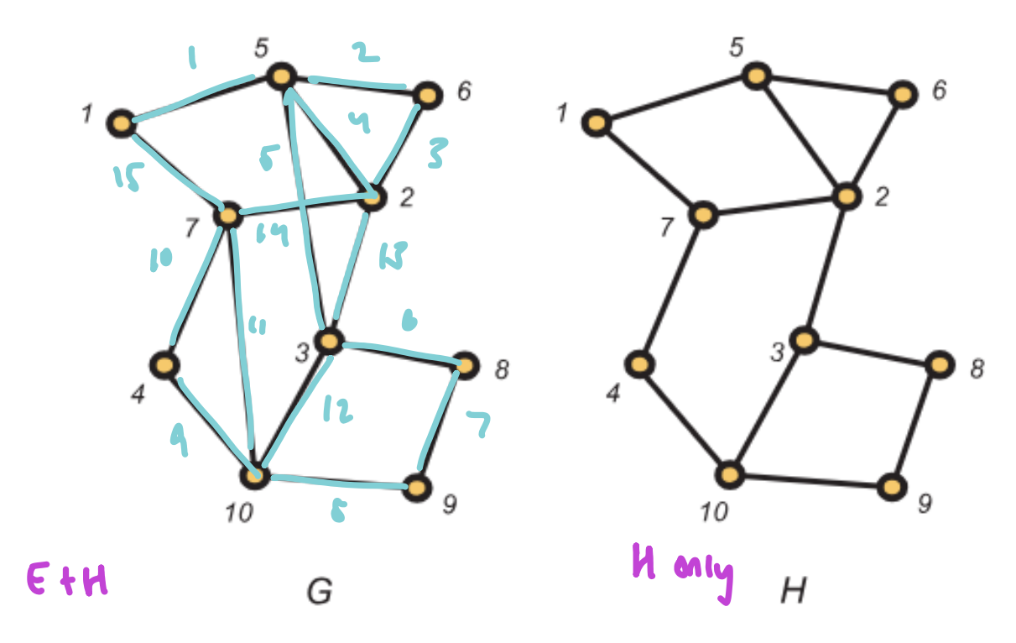
Corollary: for any graph, number of vertices of odd degree is even

Proposition: Every tree on $n \geq 2$ vertices has at least 2 leaves

Eulerian: sequence of vertices $(x_0, x_1, \ldots x_t)$ from G with repetition allowed st.
(edges)
$x_0 = x_t$, $x_i x_{i+1}$ is an edge of G, and for every edge $e \in E$ there is a unique integer i with $0 \leq i < t$ for which $e = x_i x_{i+1}$.

Thm: graph is Eulerian if and only if it is connected and every vertex has even degree.

Hamiltonian: graph $G = (V, E)$ is Hamiltonian if there exists a sequence $(x_1, x_2, \ldots x_n)$ such that every vertex of G appears exactly once, $x_1 x_n$ is an edge, and $x_i x_{i+1}$ is an edge.
(vertices)



Dirac Theorem: If G is a graph with $n \geq 3$ vertices and each vertex in G has at least $\lceil \frac{n}{2} \rceil$ neighbours, then G is hamiltonian.

Graph colouring

proper colouring: $\phi : V \to C$ st. $\phi(x) \neq \phi(y)$ if xy is an edge in G
least number of colours for proper colouring: chromatic number, $\chi(G)$

Theorem: a graph is 2-colourable if and only if it does not contain an odd cycle

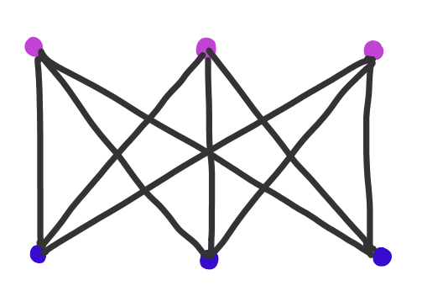
Bipartite graph: there is a partition of vertex set into two sets A and B st. subgraphs induced by A and B are independent graphs (no edges in both A or both B)

Bipartite graphs are 2-colourable.

used for 2 distinct types of objects, connections between diff. types

Complete bipartite graph: $K_{m,n}$ has vertex set $V = V_1 \cup V_2$ with $|V_1| = m$, $|V_2| = n$
Edge xy if and only if $x \in V_1$, $y \in V_2$

e.g. $K_{3,3}$



Clique: $K \subseteq V$ such that induced subgraph is complete (all vertices connected)

Clique number: $\omega(G)$, size of largest clique in G

$\chi(G) \geq \omega(G)$, where $\chi(G)$ is chromatic number
$\uparrow$ can be much larger

Generalized pigeonhole principle: If $f : X \to Y$ is a function and $|X| \geq (m-1)|Y| + 1$, then there exists an element $y \in Y$ and distinct elements $x_1, \ldots x_m \in X$ so that $f(x_i) = y$ for $i = 1, \ldots m$
element in Y with m elements mapped to it

χ and ω do not need to be close
$\downarrow$

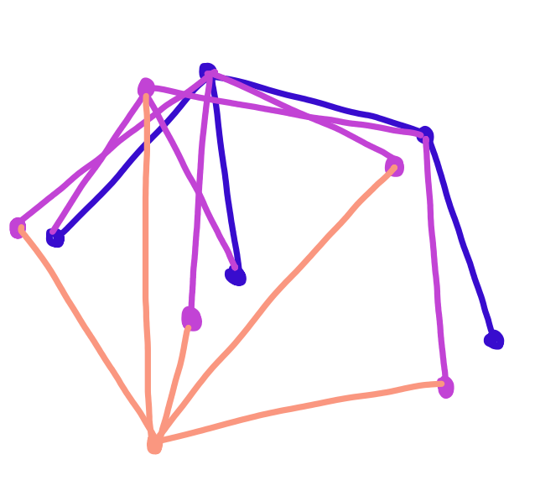
Proposition: for every $t \geq 3$, there exists a graph G_t so that $\chi(G) = t$ but $\omega(G_t) = 2$

proof: given G_t, construct G_{t+1}

let n_t be number of vertices in G : $x_1, x_2, \ldots x_{n_t}$

Add copy of G, with y_i connected to x_j iff x_i is adjacent to x_j.

Then add new vertex z adjacent to all y_i



$\omega(G_{t+1}) = 2$

$\chi(G_{t+1}) \geq t$, and $\chi(G_{t+1}) \leq t+1$ (can colour G with $\{1, \ldots t\}$, I with $t+1$, z with 1)

show: $\chi(G_{t+1}) = t+1$

contradiction: assume t colours.

WLOG, suppose z is colour t

consider nonempty set of vertices on copy of G_t to which ϕ assigns t.

??

Theorem: there exist infinitely many graphs st. $\chi(G) = \omega(G)$

Given an indexed family of sets $\mathcal{F} = \{S_\alpha : \alpha \in V\}$, we associate with $\mathcal{F}$ a graph $\mathbf{G}$ defined as follows. The vertex set of $\mathbf{G}$ is the set V and vertices x and y in V are adjacent in $\mathbf{G}$ if and only if $S_x \cap S_y \neq \emptyset$. We call $\mathbf{G}$ an ***intersection graph***. It is easy to see that every graph is an intersection graph (*Why?*), so it makes sense to restrict the sets which belong to $\mathcal{F}$. For example, we call $\mathbf{G}$ an ***interval graph*** if it is the intersection graph of a family of closed intervals of the real line $\mathbb{R}$. For example, in Figure 5.27, we show a collection of six intervals of the real line on the left. On the right, we show the corresponding interval graph having an edge between vertices x and y if and only if intervals x and y overlap.

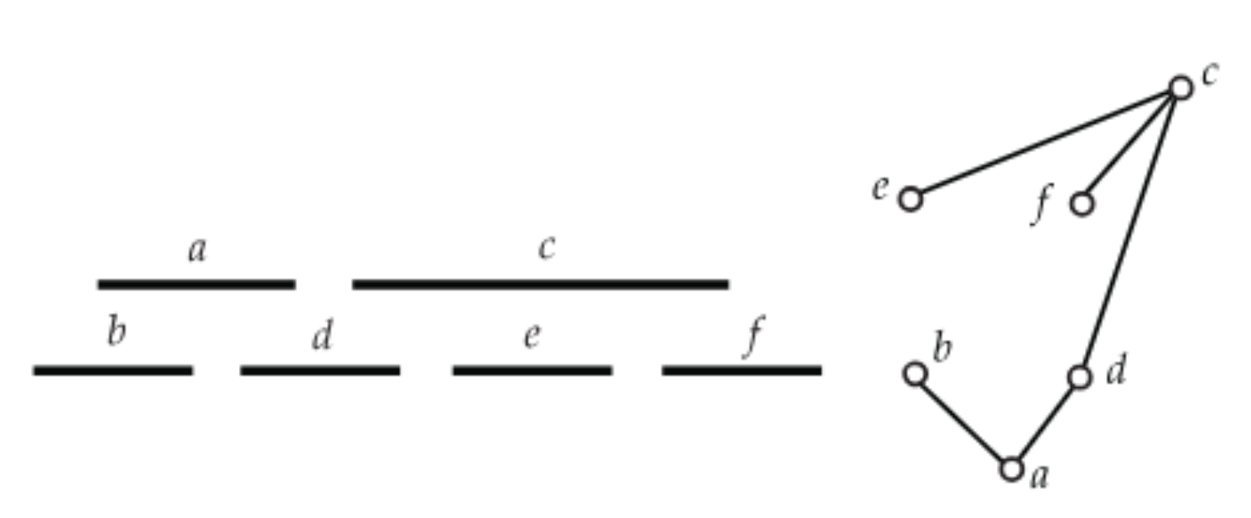


Figure 5.27. A collection of intervals and its interval graph

chromatic #
$\downarrow$
clique #

Theorem: If $G(V,E)$ is an interval graph, then $\chi(G) = \omega(G)$

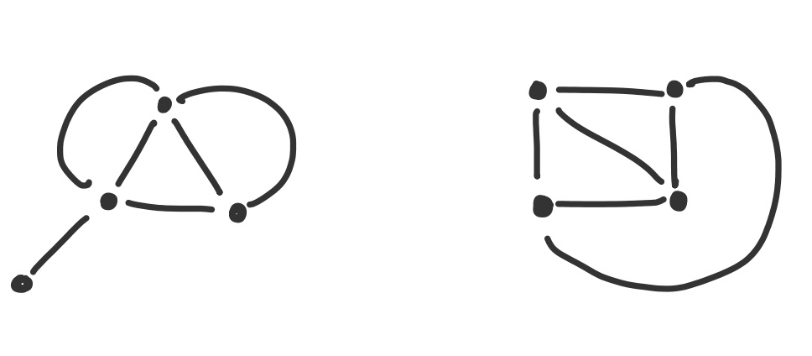
(each v_i is adjacent to at most $\omega(G) - 1$ other vertices, so for v_i, there will always be a colour in $\{1, 2, \ldots \omega(G)\}$ unused.)

A graph G is perfect if $\chi(H) = \omega(H)$ for every induced subgraph H.
Induced subgraph of interval graph is interval graph => Interval graphs are perfect

Planar graphs

Planar graph: graph drawn in $\mathbb{R}^2$ such that edges do not intersect.

e.g.



face: bounded by edges and vertices, and not containing other edges and vertices

Euler's formula: Let G be a connected planar graph with n vertices and m edges and f faces.

$n - m + f = 2$

$(v - e + f = 2)$

$\uparrow$ 0 dim $\uparrow$ 1 dim $\nwarrow$ 2 dim

Proof by induction on e:

Base case: $e = 0$, $v = 1$, $f = 1$

Consider G with e edges. Remove an edge to get G'.

Case 1: G' is connected

By IH, $v' - e' + f' = 2$

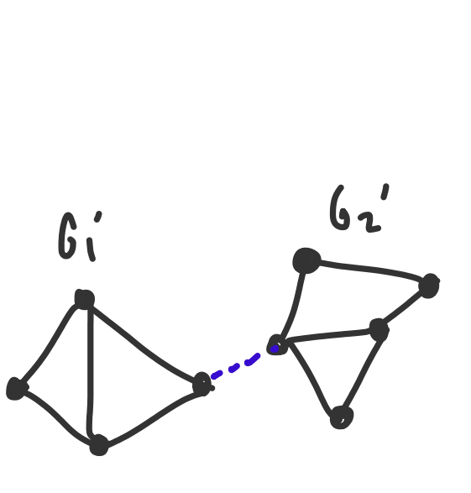
$v' = v$

$e' = e - 1$

$f' = f - 1$ (otherwise it is not connected)

Then, $v' - e' + f' = v - e + f = 2$

Case 2: G' is not connected



View G' as two connected graphs G_1' and G_2'

$v_1' - e_1' + f_1' = 2$, $v_2' - e_2' + f_2' = 2$

$\underbrace{(v_1' + v_2')}_{v} - \underbrace{(e_1' + e_2')}_{e-1} + \underbrace{(f_1' + f_2')}_{f+1} = 4$

$v - (e-1) + (f+1) = 4$

$v - e - f = 2$

Theorem: A planar graph on n vertices has at most $3n-6$ edges when $n \geq 3$
$(3v - 6 \geq e)$

Proof: count (edge, face) pairs

number of pairs $= 2e$ (each edge has inner and outer face)

each face has at least 3 edges $\Rightarrow 3f \leq 2e$

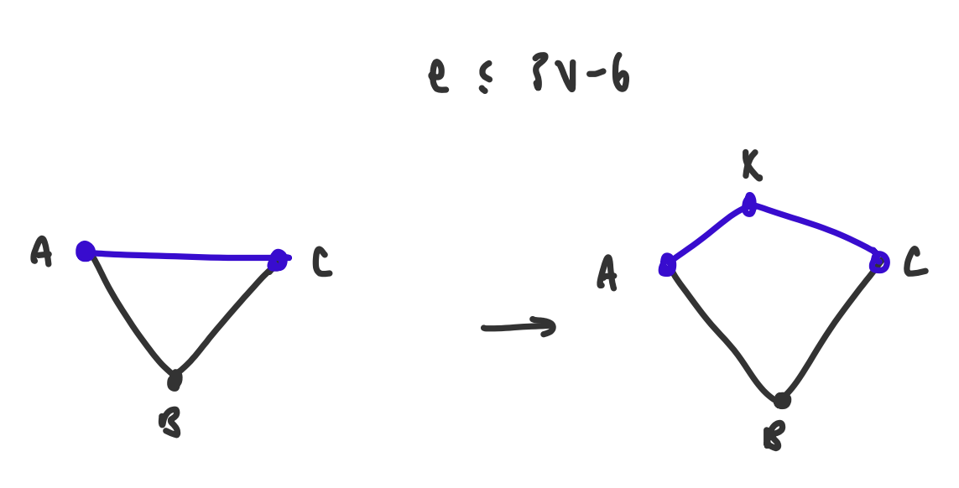
$v - e + f = 2$

$f = 2 + e - v$

$3(2 + e - v) \leq 2e$

$6 + 3e - 3v \leq 2e$

$e \leq 3v - 6$

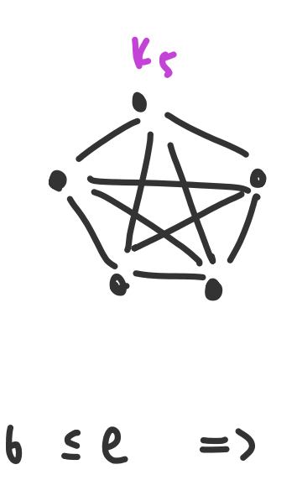


elementary subdivision: replace edge uv with new vertex uv', v'v

Kuratowski's Theorem: A graph is planar if and only if it does not contain a subgraph homeographic to K_5 or $K_{3,3}$

$\uparrow$ can check if graph is planar

Ex. Planarity of K_5



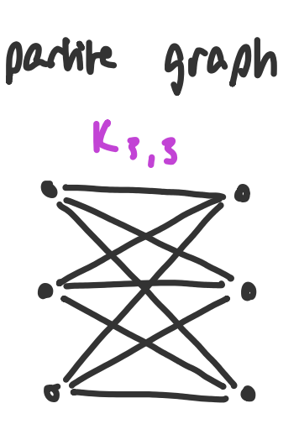
$v = 5$

$e = \binom{5}{2} = 10$

If planar, should satisfy $3v - 6 \leq e \Rightarrow 9 \leq e$

Euler's formula does not give this: " $f = 2 + e - v = 7$ if planar "

Ex. Complete bipartite graph $K_{3,3}$



$v = 6$, $e = 3 \times 3 = 9$

satisfies $3v - 6 \leq e$ $(3(6) - 6 = 12 > 9)$

Euler's formula: $f = 2 + e - v = 2 + 9 - 6 = 5$ if planar

Ex. Petersen graph is not planar